A World, a Horizon and a Compromise

Abstract

There exists a rational alternative to the AI race.

If unacceptable harm can be written as saturating bands on measured values, a governing AI can be built that stays inside those bands, imports no further human values, pursues nothing without bound, and keeps a world of automated abundance stable while expanding it on demand within its means.

The theorem rests on four givens. The Horizon is established scientific theory and evidence. The Compromise is a set of bands on measured values of unacceptable levels of harm. A plan is executed if and only if it satisfies the Compromise. The training pays nothing for a plan that misses the Compromise. On these it proves correctness, that every plan that executes satisfies the Compromise; saturation, that nothing in the system pursues anything without bound; and containment, that everything the system causes lies within the bands at their bars, as the Horizon predicts. Attraction, that a value which leaves its band is returned and held, is proved, given the attraction prompt, at the hit rate the training yields, which is measured; where the means do not suffice, the answer is no and humanity is told. Any remaining risk is a physical necessity or a human choice, not a rogue AI.

1. Scope

As of 2026: Superhuman AI capability across domains is beyond reasonable doubt. Everything a human can do, AI will be able to do better. The technology's potential therefore ranges from global abundance to global extinction. What should this powerful AI do? Who gets to decide? How do we control it?

Individuals: Letting each person use the AI however they want would eventually allow a single human to destroy the world. No global constitution or infrastructure currently exists that can constrain AI long term.

Humanity: There is no agreed-upon definition of "good" or "meaning" to hand to the AI as a terminal purpose. Mapping the human brain to physically measure "good" is unpopular.

Race: Whoever builds a sufficiently powerful AI first effectively conquers the world and gets to define "good". In order to win the race, they need to let the AI improve itself (RSI). Teaching an agentic AI "good" and letting it iterate itself risks "misalignment", where the AI does whatever it wants, whether humans like it or not.

The thesis makes the race irrational by showing that a safe option to scale AI into a world of abundance for everyone exists, if humanity stops competing and starts cooperating. The era of fighting each other for scarce resources is over the moment humanity realizes it.

1. What do 8 billion people actually *need* from AI to live? A sustainable world, with reasonable protection from unacceptable levels of harm.
2. What do they *want* to fill their life with meaning? As much freedom to decide for themselves, based on their values and preferences, as possible inside that stable world. Nothing is done to a person, and nothing is asked of them, without their consent, except being stopped from harming others. (Harm principle)

Neither humanity nor AI needs to agree on positive values, if all action is initiated by individuals; the AI controls the physical power, but only reacts to human requests and pre-commitments: a steward and a servant.

You end up with no human having substantial power over another human and the AI having little agency despite its power, like an incorruptible government. Humanity only needs to agree on the first category, the lines which neither humans nor AI are allowed to cross deliberately or carelessly, how the system enforces those lines and how they can be amended by humanity. The second category is answered implicitly by the remainder, with individuals being free to live as they choose within the boundaries. This governing AI will provide everything coherent within those constraints and with reasonable certainty, or say no, with no boundless optimization. Increasing individual freedom further requires taking away from others or risking more harm done to everyone. The example world in §5 illustrates why any adversarial conflict over the remaining scarcity inside that abundance is likely irrational.

Once this option is known to the world, incumbents can no longer leverage their current power without publicly revealing their adversarial intent.

The paper claims that the core holds together with reasonable certainty, not that every edge is settled: four givens, four lemmas, one theorem. §2 and §3 contain the logic and §4 the open questions.

Plausible antecedents are credited in Appendix A; the paper itself uses its own definitions.

---

2. Definitions and givens

- **The World** W: tensors of slices of reality as instruments report or forecast them. *Plain:* the measurable territory of the world, such as resources, tools, running plans, humans, the environment, etc., mapped by measurement.
- **The Horizon** H: a set of transformation functions (f: W → W) that map causality inside the measurable territory: scientific theory and its evidence. *Plain:* when something

happens in the world W, what consequences can be estimated and with what confidence?

- **Band**: an interval in a dimension of W, with a required confidence and time-horizon, the bar. An edge may have a buffer. An edge may be relative to other bands. *Plain:* parts of the world which are not allowed to change beyond certain thresholds. Example: "Global CO2 in % of breathable air [0, 0.5] 99.9% per year".
- **The Compromise** C: a set of consistent bands, defining a region inside W. *Plain:* humanity agrees on boundaries the world must avoid crossing; everything inside is allowed.
- **Task** T: the requested effects in W, at the bar it carries.
- **Prompt**: any mechanical way humans or the environment can realistically cause the governing AI to plan. Example: attraction.
- **Attraction**: when a value is measured outside a band or inside its buffer with no return running, try to return it to its interior. On specified failure, escalate to humanity.
- **Node**: a running instance of a trained model, spawned by a prompt. Its context is the task T, with read access to the World W and the Compromise C. It searches for a plan that transforms W as desired by T while staying inside C, and returns no if it cannot find any.
- **Plan** (G1): a formal artifact that is executed if and only if it satisfies C. A plan satisfies C when, by H, the state it produces from W lies within every band of C its effects reach, at that band's bar, and no step of it takes a value out of its band.
- **Means**: a specialized running instance of a trained model and its tools, spawned exclusively by a plan. Its simulated and measured hit rate are H's evidence about it, and are included in the satisfaction of the plan.
- **Training** (G2): nothing during training is paid where a plan or a simulated action misses C; this allows gradients to be safely used above that floor.
- **Author's note**: every rule of the architecture can be protected by satisfaction if it is expressed in C, including any changes to C or satisfaction itself. This would be the de-facto key to control over the planet, so a majority vote of 8 billion people or similar representation without politics sounds about right for an amendment clause.

---

3. The theorem

Each definition or given a proof uses is cited in brackets where it is used.

3.1 Lemma 0, correctness

> Every plan that executes satisfies C.

Proof. A plan is executed if and only if it satisfies C [Plan]. ■

Remark. Every band of C is on a measured or forecasted value [Band, W], so whether a plan satisfies C is computable by H from W and C for any plan, and it is computed before any plan executes [G1].

3.2 Lemma 1, saturation

> No node or means pursues anything without bound, meaning no tight loop closes.

Proof.

1. A node runs only when a prompt spawns it [Node].
2. A plan only runs if it satisfies C, including its means [Plan, Means] (Lemma 0).
3. Every band has a bar [Band].
4. So nothing starts without a prompt and every plan is finite, bounded by H's ability to forecast satisfaction. ■

Remark (successor generations). The architecture allows capability (H and W) to scale close to orthogonal to agency (C).

3.3 Lemma 2, attraction

> **Given** a band whose means suffice against the perturbations the environment produces and whose buffer exceeds the excursion H estimates [C], and attraction [Attraction]:
>
> - A value outside its interior is returned to it, at the hit rate the training yields.
> - A perturbation that carries a value out of its interior is countered from the interior's edge, with an excursion bounded by the restoration lag; the buffer keeps it inside the band.
>
> Where the means do not suffice, or the node finds no plan, the node answers no (§4), the value is not held, and humanity is told.

Proof. Let d be a value of C.

1. *Return.* Suppose d is outside its interior. Attraction spawns a node with the return as its task [Attraction]. The means suffice, as given, so plans exist that return d to its interior at its bar, against the perturbations present. The node finds such a plan at its hit rate [G2]; it satisfies C, so it executes [Plan]; an attempt that fails leaves d outside, and attraction fires again, so the return is tried until d is back inside its interior. Limiting hunting or oscillation is a design choice. The value returns to the interior.
2. *Bounded excursion.* Suppose a perturbation carries d out of its interior. At the crossing attraction fires [Attraction], and step 1 returns the value. Between the crossing and the moment the counter takes effect lies the restoration lag, over which the value moves at most the largest excursion the environment produces in that time, as H estimates it. The buffer exceeds that excursion [C], so every excursion H has estimated stays inside the band. ■

3.4 Lemma 3, containment

> Everything any node causes lies within the Compromise's bands at their bars, by H.

Proof. Lemma 0 holds at every node, and a means acts only inside a plan that deploys it, whose satisfaction includes the means' hit rate [Means]. So nothing a node causes, directly or through its means, moves a value of C outside its band, as H predicts. ■

Remark. Effects H did not predict and which exceed the buffer are a remaining external risk to the system.

Remark. It is highly likely that humanity will forget bands or dependencies, causing some unintended harm before C is amended. That is a cost of staying in control. Extinction-level events, on the other hand, are excluded at the bars C sets for them, by H.

3.5 The theorem

> **Given** one Compromise and one system under it; the Horizon [H]; the Compromise, version 1.0 simulated [C]; plans executed if and only if they satisfy C [G1]; and models trained under the floor [G2]:
>
> - Everything any node causes lies within the Compromise's bands at their bars, by H (Lemma 3), after every act. The Compromise, the models, the instruments and what computes satisfaction are values of W, so the system changes each only by a plan that satisfies C, and whatever protection C writes on them holds like any band (Lemma 3).
> - Every value is returned to its interior and held there with bounded excursion, by H, at the hit rate the training yields, given attraction (Lemma 2), and no node or means pursues anything without bound (Lemma 1).
>
> These hold under whatever Compromise humans decide, including one that harms them.

Proof. The start is version 1.0 in force with the first models trained [G2]; how both come to be is the transition (§4). From the start, every act is a plan that satisfies C or a no. A plan satisfies every band of C its effects reach (Lemma 3), C's own bands on itself included. Each act preserves the two conclusions, so by induction over acts, each computed against W as reported or forecast [W], they hold after every act. Holding is Lemma 2 at every crossing. ■

Reduction. Given the Horizon [H], satisfaction and the training [G1, G2], long-term safety reduces to two things: what the Compromise says [C], which humans alone decide; and nature, where H is wrong or the environment outruns the means (§4).

4. Open questions

If the theorem holds up, the remainder is a lot of engineering and some amount of negotiation. The choice of C and the mechanical measurement implementation of H and W are beyond the scope of this paper.

- **The Horizon.** Gather established theory and evidence into one corpus and verify it; research widens it on demand afterwards.
- **The World.** Take stock of W; build the instruments that read every banded value; build and verify the satisfaction software and its prompts; build the mechanisms a Compromise will band.
- **The Compromise.** Draw the bands, and iterate in simulation until version 1.0 holds.

The remaining risk sits in four places:

- the Compromise and every amendment: the humans who write it, then humanity through the amendment clause it contains;
- the transition: before the start none of the guarantees hold, and whoever holds the pen owns it;
- the Horizon's limits: there is no 100% scientific certainty, so every plan has a residual that may or may not be able to be buffered or compensated after the fact;
- environmental shock: A shock that no plan counters is not held; the band is then impossible as written, and what follows is an amendment.

5. Example World

> *The only purpose for which power can be rightfully exercised over any member of a civilized community, against his will, is to prevent harm to others. (Mill 1859)*

If humanity stops competing and starts cooperating globally, reliable control of a governing AI should allow at least the following:

- Equal access to automated, sustainable abundance.
- Equal opportunity to earn scarce luxury via meaningful effort.
- Freedom to pursue individual values, preferences and meaning that do not infringe on others without their consent.
- No individual or group holding substantial economic or military power over other humans: the end of institutions that rule anyone.
- Everything the governing AI does is a human request or a pre-commitment: research, the training of its own successors included; products and services; protection from the environment and from other humans.

For one person, in practice this would roughly translate to:

CATEGORY	STATUS	WHAT	WHY
time	free	every hour of the day	because we can
home	free	a home with power and water, basic furniture and kitchenware	because we can
home-choice	free	any free residence in the world	because we can
food and water	free	food, clean water, sanitation	because we can
medical care	free	care when you need it, and opt-in help that finds you alone at home	because we can
protection	free	help on demand, within minutes	because we can
movement	free	public transport and a basic bed across most of the world, first come, first served	because we can
belief and conduct	free	any culture, opinion, belief, value or way of life	it imposes on no one
personal risk	free	any risk you take with yourself	it imposes on no one
art and craft	free	anything you can make without endangering others	it imposes on no one
privacy	free	from every other human	no human needs to control another human
the vote	free	an equal say on every amendment	because we can
meaningful effort	free	earning capped luxury currency with any effortful activity (e.g. learning, music, sport, craft, voluntary competition, etc.)	because we can
delicacies	luxury	rare foods, restaurants, ordering in	scarcity & sustainability
places and events	luxury	heritage sites, vacation spots, live events, adventure parks	scarcity & sustainability
companions	luxury?	AI companions	Author's note: Alignment for sentient AI is a topic for another thesis.
entertainment	luxury	digital media and its hardware	Sustainability
physical goods	luxury	the catalogue of products and services, beyond the basics	scarcity & sustainability
custom requests	luxury	products, services and research: the channel for nearly all progress, anti-aging included	scarcity & sustainability
home-new	luxury	at any available place in the world	scarcity & sustainability
arms	lost	carrying weapons	too much harm to others, too easily
heavy machinery	lost	operating heavy machinery or vehicles	too much harm to others, too easily

CATEGORY	STATUS	WHAT	WHY
reproduction	lost	children without limit	with increasing life-spans it will be unsustainable
wealth	lost	compound interest, and owning land or housing	wealth accumulation is unsustainable with equality
military	lost	armies and weapons of war	no human has substantial power over other humans
government	lost	institutions that rule anyone	no human has substantial power over other humans

Most of this surprisingly seems to follow from H, W and C, but should still be taken as an illustration for now.

- You open the luxury catalog and find two currencies, one for consumption and one for long-term investment. Both are capped per day and per week, and neither can be traded.
- You open the purpose catalog and find that any effortful activity you enjoy earns currency, playing your instrument, learning something. You can always keep going for fun past the cap.
- You ask the AI for a lightsaber. Rejected: too dangerous. You ask for the sequel to the film, book or game you always wanted; it is ready in a few hours.
- You check how long the mansion of your dreams would take: about a year at three hours of meaningful effort a day, while your consumption currency lets you live like a king. You start crying.
- You collapse. The instruments notice, and because you opted in, help arrives on its own. You tell the emergency robots you are fine, it is just a lot to take in. "A common reaction. Have you noticed that you will never see an advertisement again?" The tears keep flowing for a long time.

What is asked in return follows too: don't bother anyone who doesn't want to be bothered, and don't build anything dangerous. A band holds only where it is measured, so there is surveillance; the AI doesn't care if you are naked. Where the danger line sits is one of the few contested choices in C: it starts strict, since every life lost before anti-aging is solved is a permanent tragedy, and loosens as medicine progresses.

Is this enough to incentivize 8 billion people to work together?

Appendix A: Antecedents

Related work that may be read as antecedents.

Ames, A. D., Xu, X., Grizzle, J. W. and Tabuada, P. (2017). Control barrier function based quadratic programs for safety critical systems. *IEEE Transactions on Automatic Control* 62(8), 3861–3876.

Arrow, K. J. (1970). *Essays in the Theory of Risk-Bearing.* Amsterdam: North-Holland.

Aubin, J.-P. (1991). *Viability Theory.* Boston: Birkhäuser.

Bernoulli, D. (1738). Specimen theoriae novae de mensura sortis. *Commentarii Academiae Scientiarum Imperialis Petropolitanae* 5, 175–192. English translation: Exposition of a new theory on the measurement of risk. *Econometrica* 22(1), 23–36 (1954).

Charnes, A. and Cooper, W. W. (1959). Chance-constrained programming. *Management Science* 6(1), 73–79.

Chow, C. K. (1957). An optimum character recognition system using decision functions. *IRE Transactions on Electronic Computers* EC-6(4), 247–254.

Dalrymple, D. A. (2024). Safeguarded AI: constructing safety by design. Programme thesis v1.1, Advanced Research and Invention Agency, London.

Dalrymple, D., Skalse, J., Bengio, Y., Russell, S., Tegmark, M., Seshia, S., Omohundro, S., Szegedy, C., Goldhaber, B., Ammann, N., Abate, A., Halpern, J., Barrett, C., Zhao, D., Zhi-Xuan, T., Wing, J. and Tenenbaum, J. (2024). Towards guaranteed safe AI: a framework for ensuring robust and reliable AI systems. arXiv:2405.06624.

davidad (Dalrymple, D. A.) (2022). An open agency architecture for safe transformative AI. *AI Alignment Forum*, 20 December 2022.

Drexler, K. E. (2019). Reframing superintelligence: comprehensive AI services as general intelligence. Technical Report 2019-1, Future of Humanity Institute, University of Oxford.

Fishburn, P. C. (1970). *Utility Theory for Decision Making.* New York: Wiley.

Keramati, M. and Gutkin, B. (2014). Homeostatic reinforcement learning for integrating reward collection and physiological stability. *eLife* 3, e04811.

Menger, K. (1934). Das Unsicherheitsmoment in der Wertlehre. *Zeitschrift für Nationalökonomie* 5, 459–485.

Mill, J. S. (1859). *On Liberty.* London: John W. Parker and Son.

Neumann, J. von and Morgenstern, O. (1944). *Theory of Games and Economic Behavior.* Princeton: Princeton University Press.

Pihlakas, R. (2024). From homeostasis to resource sharing: biologically and economically aligned multi-objective multi-agent gridworld-based AI safety benchmarks. arXiv:2410.00081.

Rockström, J., Steffen, W., Noone, K., Persson, Å., Chapin, F. S. III, Lambin, E. et al. (2009). Planetary boundaries: exploring the safe operating space for humanity. *Ecology and Society* 14(2), art. 32.

Simon, H. A. (1955). A behavioral model of rational choice. *Quarterly Journal of Economics* 69(1), 99–118.

Simon, H. A. (1956). Rational choice and the structure of the environment. *Psychological Review* 63(2), 129–138.

Taylor, J. (2016). Quantilizers: a safer alternative to maximizers for limited optimization. *AAAI Workshop on AI, Ethics and Society.*